

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Kind: `claude`  
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xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-  
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- SHA-256: `9983852d2f1595eef0c7784ef39d653bda7212f19f5b364113fe3e25df660a74`  
- Source modified: `2026-06-06T17:30:14+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:28:43.700Z)

Web Searcher: broad/primary

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Your angle: ****broad/primary**** ? Maps the overall landscape of method families and accuracy baselines for racket-sport rally/point boundary detection ? the primary framing the user needs to rank approaches.

Search query: `automatic rally point segmentation badminton tennis video temporal action localization methods survey`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-06T17:28:46.600Z)

I'll research this question. Let me start with the primary search.

3. assistant (2026-06-06T17:28:49.315Z)

I have the search tools loaded. Let me run several searches in parallel to cover the landscape.

4. user (2026-06-06T17:28:57.390Z)

Web search results for query: "ShuttleSet badminton dataset rally stroke shuttle trajectory license"

Links: [{"title": "(PDF) ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://www.researchgate.net/publication/371413847_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/pdf/2306.04948"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining", "url": "https://dl.acm.org/doi/10.1145/3580305.3599906"}, {"title": "[PDF] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/ShuttleSet:-A-Human-Annotated-Stroke-Level-Singles-Wang-Huang/405132e19aa07c8a9ae6f8a4e6f440b00a105027"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Request PDF", "url": "https://www.researchgate.net/publication/372910678_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}, {"title": "[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/abs/2306.04948"}, {"title": "(PDF) ShuttleSet22: Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset", "url": "https://www.researchgate.net/publication/371909519_ShuttleSet22_Benchmarking_Stroke_Forecasting_with_Stroke-Level_Badminton_Dataset"}, {"title": "[2306.15664] ShuttleSet22: Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset", "url": "https://arxiv.org/html/2306.15664"}, {"title": "ShuttleSet22: Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset", "url": "https://arxiv.org/html/2306.15664v1"}]

Based on the search results, here's what I found about the ShuttleSet badminton dataset:

Overview

ShuttleSet is the largest publicly-available badminton singles dataset with annotated stroke-level records. It contains 36,492 strokes, 3,685 rallies, and 104 sets from 44 matches between 2018 and 2021 played by 27 high-ranking men's and women's players.

Annotation and Features

ShuttleSet is manually annotated with a computer-aided labeling tool to increase the labeling

efficiency and effectiveness of selecting the shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke. The dataset includes rally category information, temporal data about hitting times and frames, and other stroke-level metadata.

Related Work

The search results also mention ShuttleSet22, which is collected from high-ranking matches in 2022 and consists of 30,172 strokes in 2,888 rallies in the training set, 1,400 strokes in 450 rallies in the validation set, and 2,040 strokes in 654 rallies in the testing set with detailed stroke-level metadata within a rally.

However, the search results do not contain specific information about the license terms for the ShuttleSet dataset. If you need details about the licensing, you may want to check the official paper or dataset repository directly.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-06T17:28:59.044Z)

Web search results for query: "automatic rally point segmentation badminton tennis video temporal action localization methods survey"

Links: [{"title": "(PDF) A Survey on Video Action Recognition in Sports: Datasets, Methods and Applications", "url": "https://www.researchgate.net/publication/366600156_A_Survey_on_Video_Action_Recognition_in_Sports_Datasets_Methods_and_Applications"}, {"title": "A Survey on Video Action Recognition in Sports - GitHub Pages", "url": "https://xhyccc.github.io/1.pdf"}, {"title": "A Survey on Video Action Recognition in Sports: Datasets, ...", "url": "https://scispace.com/pdf/a-survey-on-video-action-recognition-in-sports-datasets-qakudotj.pdf"}, {"title": "ProTAL: A Drag-and-Link Video Programming Framework for Temporal Action Localization | Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems", "url": "https://dl.acm.org/doi/10.1145/3706598.3713741"}, {"title": "Action Spotting and Precise Event Detection in Sports: Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v1"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "SmartTennisTV: Automatic indexing of tennis videos", "url": "https://arxiv.org/pdf/1801.01430"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "(PDF) Analyzing Racket Sports From Broadcast Videos", "url": "https://www.academia.edu/70243184/Analyzing_Racket_Sports_From_Broadcast_Videos"}]

I found relevant search results about automatic rally point segmentation, temporal action localization in sports videos (particularly badminton and tennis), and related methods. Here's what the search revealed:

Survey and Methods

Badminton Olympic dataset can be used for multiple tasks including player detection, point localization, action recognition and localization. The dataset includes annotations of time boundaries and labels of strokes with 12 categories of strokes, such as serve and lob.

Rally Segmentation Approaches

****Tennis Rally Segmentation:**** One approach uses a lightweight, purely audio-based detection model that trains a neural network to identify the distinct acoustic signatures of racquet-ball impacts from the broadcast audio track. The hitting moments are then clustered in the temporal domain to form candidate rally intervals, and specific filtering rules are applied to retain valid segments corresponding to complete, uninterrupted rallies.

Another method for tennis videos observes that in broadcast tennis videos, the camera is only

overhead when the rally is in play, with the background mostly static after the serve begins and remaining the same until a player wins a point.

****Badminton Rally Segmentation:**** Broadcast badminton matches are divided into gameplays and replays, where a "rally" frame is defined as a frame where actual gameplay is happening and a "non-rally" frame is where replays or player resting occurs.

Related Datasets

The P2A table tennis dataset comprises 2,721 untrimmed broadcast videos with 272 hours total length, featuring professional annotations with 14 fine-grained stroke classes, with each action annotated with precise start and end frames.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:29:02.212Z)

Good results. Let me dig into the more specific method families: audio-based hit detection, the broadcast badminton rally segmentation paper, and VLM distillation for sports.

7. user (2026-06-06T17:29:11.360Z)

Web search results for query: "badminton rally detection court homography player pose serve detection state machine broadcast"

Links: [{"title": "Service and End of Rally Detection in Badminton Videos | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "ML Beginner's Guide For Badminton Pose Estimation Model", "url": "https://www.labelerr.com/blog/ml-beginners-guide-for-badminton-pose-estimation-trajectory-model/"}, {"title": "Vid2Player: Controllable Video Sprites that Behave and Appear like Professional Tennis Players", "url": "https://arxiv.org/pdf/2008.04524"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/pdf/2502.21085"}, {"title": "Court to conversation: Tactical badminton analysis via computer vision and RAG-enhanced LLMs - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/abs/pii/S0950705125020659"}, {"title": "GitHub - RichardPinter/badminton_shot_type", "url": "https://github.com/RichardPinter/badminton_shot_type"}]

I've performed the web search for information about badminton rally detection, court homography, player pose, serve detection, state machine, and broadcast video analysis.

Here are the key findings:

****Rally Detection and Serve Detection:****

Techniques have been developed to detect the start of service and end of rally, utilizing court details and poses to accurately identify the service action from other badminton actions. A rally is a sequence of strokes that start from a serve and ends when the shuttlecock falls on the court and a point is awarded to a player.

****Court Homography:****

Using homography techniques, 2-dimensional broadcast view camera coordinates are mapped to top view court coordinates by computing a homography matrix between court corners and top view corners.

****Player Pose Estimation:****

Computer vision pipelines use object detection (YOLO) and pose estimation (MediaPipe) to generate structured datasets of spatio-temporal events.

****State Machine Approach:****

The shot cycle forms a simple state machine that governs each player's behavior during a rally, with both players repeatedly executing new shot cycles.

****Broadcast Video Analysis:****

Game rules and players' spatial positions on the court can be exploited for automatically predicting point outcomes, by recording predicted players' positions and strokes played in each frame.

A practical implementation pipeline combines these elements: Court calibration via homography matrix, TrackNetV3 shuttlecock detection, MMPose player pose estimation, data collation, and stroke prediction using transformer models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-06T17:29:12.760Z)

Web search results for query: "audio-based tennis racket ball hit detection rally segmentation acoustic shot detection deep learning"

Links: [{"title": "(PDF) Detection of ball hits in a tennis game using audio and visual information", "url": "https://www.researchgate.net/publication/261119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis", "url": "https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"}, {"title": "Detection of Ball Hits in a Tennis Game Using Audio and Visual Information", "url": "http://www.apsipa.org/proceedings_2012/papers/316.pdf"}, {"title": "Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/"}, {"title": "[PDF] Detection of Tennis Events from Acoustic Data | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/Detection-of-Tennis-Events-from-Acoustic-Data-Baughman-Morales/5757cc038c864dc0f2d676794faaa4ff8961a97f"}, {"title": "(PDF) Detection of Tennis Events from Acoustic Data", "url": "https://www.researchgate.net/publication/336663325_Detection_of_Tennis_Events_from_Acoustic_Data"}, {"title": "Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11843912/"}]

Based on the search results, here's what I found regarding audio-based tennis racket ball hit detection, rally segmentation, acoustic shot detection, and deep learning:

Key Research Areas

Audio-based detection models have been employed to identify the distinct acoustic signatures of racquet-ball impacts from broadcast audio tracks, with applications in automating rally analysis.

Machine Learning Approaches

Machine learning algorithms have been developed to classify the sound of the ball's impact on the racket and the ground to measure the time between them and give immediate feedback to the player. Several machine learning techniques including Decision Tree, Random Forest, SVM, and XGBoost have been investigated for classifying sounds from tennis matches.

Deep Learning Methods

A real-time-capable pipeline combining energy-based peak detection with convolutional neural networks trained on a dataset of 3,396 bounce samples across 10 racket configurations has been proposed.

Rally Segmentation Applications

Professional tennis broadcasts are segmented into individual rally clips by organizing the dataset at the rally level from full match broadcasts. Training data is augmented by overlaying various background sounds such as crowd cheering, clapping, and announcer voices to ensure the model is robust against complex ambient stadium noises.

Challenges

Distinguishing a ball sound from noise is challenging when background noise coincides with the shot, and even in situations with high noise levels, the model should recognize and separate the ball sound, necessitating strong feature extraction and classification methods.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:29:15.826Z)

These are highly relevant. The "Service and End of Rally Detection in Badminton Videos" paper directly addresses pose/court-based rally segmentation. Let me run two more searches on VLM distillation for video and the broadcast badminton rally segmentation specifics, plus TrackNet licensing.

10. user (2026-06-06T17:29:22.610Z)

Web search results for query: "video large language model teacher distillation temporal action localization student model weakly supervised sports"

Links: [{"title": "Can MLLMs Guide Weakly-Supervised Temporal Action Localization Tasks?", "url": "https://arxiv.org/html/2411.08466v1"}, {"title": "Event-Oriented State Alignment Network for Weakly Supervised Temporal Language Grounding - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11431080/"}, {"title": "Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models", "url": "https://arxiv.org/html/2411.08466v2"}, {"title": "[2411.08466] Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models", "url": "https://arxiv.org/abs/2411.08466"}, {"title": "Weakly Supervised Temporal Action Localization via Representative Snippet Knowledge Propagation | Request PDF", "url": "https://www.researchgate.net/publication/363908213_Wweakly_Supervised_Temporal_Action_Localization_via_Representative_Snippet_Knowledge_Propagation"}, {"title": "Weakly Supervised Temporal Action Localization via Dual-Prior Collaborative Learning Guided by Multimodal Large Language Models", "url": "https://arxiv.org/pdf/2411.08466"}]

I found relevant research on your query. Here are the key results:

The most recent research introduces a learning paradigm called MLLM4WTAL that harnesses the potential of Multimodal Large Language Models (MLLMs) to offer temporal action key semantics and complete semantic priors for conventional Weakly-supervised Temporal Action Localization (WTAL) methods.

Recent breakthroughs in MLLMs have gained significant recognition within the deep learning community, where the fusion of Video Foundation Models (VFMs) and Large Language Models (LLMs) has proven instrumental in constructing robust video understanding systems. However, their operation demands substantial memory and computational resources, underscoring the continued importance of traditional models in video comprehension tasks.

Additionally, research shows that a classification-based teacher model is typically employed to generate pseudo labels, which are then used to train a regression-based student model for precise boundary prediction in temporal action localization tasks.

The search results include papers from November 2024 and June 2025 that address how Large Language Models can guide weakly-supervised temporal action localization, along with discussions of knowledge distillation approaches between teacher and student models for this task.

REMINDER: You MUST include the sources above in your response to the user using markdown

hyperlinks.

11. user (2026-06-06T17:29:24.267Z)

Web search results for query: ""Towards Real-Time Analysis of Broadcast Badminton Videos" rally segmentation rally non-rally classification accuracy"

Links: [{"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos | Request PDF", "url": "https://www.researchgate.net/publication/325002967_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "Badminton Video Analysis based on Spatiotemporal and Stroke Features", "url": "https://www.researchgate.net/publication/317159692_Badminton_Video_Analysis_based_on_Spatiotemporal_and_Stroke_Features"}, {"title": "[2308.12199] Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/abs/2308.12199"}, {"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition | Request PDF", "url": "https://www.researchgate.net/publication/388099044_VideoBadminton_A_Video_Dataset_for_Badminton_Action_Recognition"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "AI-powered Badminton Video Detection: Enhancing Gameplay Analysis and Training | TechRxiv", "url": "https://www.techrxiv.org/doi/full/10.36227/techrxiv.23708325.v2"}]

I found relevant information about the paper "Towards Real-Time Analysis of Broadcast Badminton Videos" and related research on badminton video analysis. Here are the key findings:

Point Segmentation and Rally Classification

A "rally" frame is defined as a frame where the actual gameplay is happening and a "non-rally" frame as a frame where the replay (highlights) has been shown, or players are resting, etc. This classification allows researchers to keep all the rally frames, group them into rallies according to the points scored by the players and discard the non-rally frames.

Classification Accuracy Results

On 10 Olympic matches with 20 players, a research framework achieved 95.44% point segmentation accuracy, which refers to rally/non-rally classification performance.

Additionally, related research on rally detection shows strong results: The BiLSTM model achieved the highest classification accuracy of approximately 95% for detecting rally starts, with frame intervals of 4, while the CNN achieved similar accuracy at frame intervals of 5.

Dataset Annotation

The annotation effort included 23,156 frames in total, where 15,147 were non-rally frames and 8,009 were rally frames.

These results demonstrate that modern deep learning approaches can effectively segment broadcast badminton videos to distinguish between rally and non-rally segments with high accuracy for downstream analysis tasks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-06T17:29:30.130Z)

I have strong coverage now. Let me verify two key items: the "Service and End of Rally Detection" paper (directly on-topic for pose/court state machines) and the SN Computer Science rally-detection comparative analysis. I have enough to rank. Let me confirm the licensing detail

on TrackNet/ShuttleSet with one quick fetch to ground the dataset-license claim, since that's load-bearing for the user.

13. user (2026-06-06T17:29:32.958Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15

Redirect URL: https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15

Status: 303 See Other

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://idp.springer.com/authorize?response_type=cookie&client_id=springerlink&redirect_uri=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-031-69073-0_15"

- prompt: "Summarize the method for service start and end-of-rally detection in badminton videos. What inputs does it use (pose, court homography, etc.)? What accuracy/F1 does it report? Does it run locally without cloud models?"

14. assistant (2026-06-06T17:29:42.819Z)

That's an auth redirect, I'll skip it. I have enough from the search snippets to rank the results confidently. Let me compile the final structured output, ranked by relevance to the original question (local-first, permissive, single-GPU rally segmentation; with emphasis on audio, pose/court state machines, datasets/licenses, and VLM distillation).

15. user (2026-06-06T17:30:02.117Z)

Structured output provided successfully

16. assistant (2026-06-06T17:30:13.974Z)

Structured output delivered. I returned 6 ranked results covering the primary method-family landscape for racket-sport rally/point segmentation, prioritized for the user's local-first, permissive, single-GPU constraints:

1. Service/End-of-Rally Detection in Badminton (pose + court state machine, fully local) ? high
2. Real-Time Broadcast Badminton Analysis (rally/non-rally classification, ~95% baseline) ? high
3. ShuttleSet dataset (rally-level labels to fix the n=2 data-starvation problem) ? high
4. ML acoustic hit-detection (the audio family they already have inputs for) ? high
5. MLLM4WTAL (validates/refines the cloud-teacher ? local-student distillation question) ? high
6. Action Spotting & Precise Event Detection survey (broad TAL framing) ? medium

One caveat flagged in the output: ShuttleSet's commercial-license terms need verification from the repo (the Springer rally-detection paper was behind an auth wall, so I grounded that family from search snippets rather than full text).